Project Synopsis

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- **Title**
  - AI-Powered Personal Finance Assistant: A Full-Stack Analytics Platform with Machine Learning and NLP Chatbot for Intelligent Financial Decision Making

- **Problem Statement (200 words)**
  - In today's fast-paced digital era, personal financial management remains one of the most neglected yet critical aspects of everyday life, particularly among students, young professionals, and middle-income households. Despite the widespread availability of banking apps and budgeting tools, the majority of individuals lack access to an intelligent, personalized system that can understand their unique financial behaviour and provide actionable, data-driven guidance. Existing financial management solutions either require expensive subscriptions, are overly complex for non-finance users, or simply lack AI-driven intelligence that adapts to individual spending habits. There is a critical gap in the market for a free, accessible, full-stack web application that integrates real-time expense tracking, machine learning-based prediction, anomaly detection, and a conversational AI chatbot — all in one unified platform. This project proposes the development of an AI-Powered Personal Finance Assistant that addresses these challenges by enabling users to log transactions, visualize spending patterns through interactive dashboards, receive ML-based predictions on future expenses, and interact with an NLP chatbot to get personalized financial advice in natural language. The system is built on the MERN stack (MongoDB, Express.js, React.js, Node.js) with a Python-based AI/ML microservice, making it both scalable and technologically comprehensive.

- **Objectives of the Project (3objectives)**
  - To design and develop a full-stack web application using the MERN stack that enables users to record, categorise, and visualise their personal income and expenses in real time, providing a comprehensive financial dashboard with interactive charts and budget tracking features.
  - To implement Artificial Intelligence and Machine Learning models including Linear Regression, LSTM (Long Short-Term Memory) for expense forecasting, Isolation Forest for anomaly detection, and a Naive Bayes classifier for

automatic transaction categorisation, integrated via a Python Flask microservice API.

- **Project Methodology (500 words)**
  - This project follows an Application-Based methodology, combining agile development practices with data science workflows. The development is divided into three interconnected layers: a full-stack web application, an AI/ML backend service, and a business intelligence reporting layer. The frontend is developed using React.js with Tailwind CSS, providing a responsive single-page application where users can register, log transactions, view dashboards, and chat with the AI assistant. The backend REST API is built with Node.js and Express.js, handling authentication (JWT), transaction CRUD operations, and routing requests to the ML microservice. MongoDB serves as the primary database via Mongoose ODM. The AI/ML service is developed in Python using Flask as the API framework. It exposes three endpoints: (1) expense prediction using LSTM and Linear Regression trained on historical transaction data, (2) anomaly detection using the Isolation Forest algorithm which flags unusual spending in real time, and (3) an NLP chatbot endpoint that uses intent classification and integrates with an LLM API to generate personalised financial responses based on the user's actual financial data fetched from MongoDB. Type of Project: Application-Based. Data Collection Methods: Secondary data from Kaggle's Personal Finance Dataset for model training; Primary data collected live through user input via the application interface.
  - The project will not incorporate real-time bank account integration or direct API connections to financial institutions such as UPI gateways or net banking systems, as this would require RBI compliance, financial licensing, and data security certifications beyond the scope of an academic project. The ML models are trained on generalised public datasets and may not perfectly reflect hyper-localised Indian spending patterns such as festival-season spikes or region-specific cost-of-living variations. Model accuracy is dependent on the volume and quality of training data, and predictions may carry a margin of error, particularly for first-time users with limited transaction history. The NLP chatbot is limited to predefined financial intents and may not handle highly complex or ambiguous queries outside its training scope. Multi-currency support and investment portfolio management (stocks, mutual funds, SIPs) are

outside the scope of this version of the application. The system is designed as a personal finance tool and does not address business accounting or enterprise financial management use cases.

- The Work Plan below outlines the key activities to be completed across 8 weeks of the project, covering requirement analysis, development, AI/ML integration, testing, and final submission.

- **Limitation (200 words)**
  - Provide an understanding of the subject knowledge by specifying aspects of the problem that will not be addressed in the project.

- **Work Plan (Week 1 to Week 8)**
  - The following week-wise plan outlines the structured development and research activities for the AI-Powered Personal Finance Assistant project over 8 weeks.

Week No.	Activities Completed
Week 1	a) Project requirement analysis and literature review on personal finance and AI/ML techniques b) System architecture design: MongoDB schema, REST API endpoints, React component hierarchy c) GitHub repository setup, project folder structure initialisation, MongoDB Atlas configuration
Week 2	a) Node.js and Express.js backend setup: user authentication APIs using JWT and bcrypt b) MongoDB Mongoose schema design for User, Transaction, and Category collections c) REST API development: CRUD endpoints for transaction management and analytics summary
Week 3	a) React.js frontend setup with Tailwind CSS: login, register, and dashboard pages b) Transaction management UI: add, edit, delete income and expense entries with category filters c) Frontend-backend integration using Axios: connect React app to Node.js REST APIs
Week 4	a) Python Flask microservice setup; data preprocessing using Pandas on Kaggle finance dataset

	b) Linear Regression and LSTM model development for monthly expense prediction by category c) Isolation Forest anomaly detection model training and Flask API endpoint creation
Week 5	a) NLP chatbot development: intent classification model and LLM API integration for personalised responses b) Chatbot context-aware responses: fetch live user financial data from MongoDB for AI prompts c) Transaction auto-categorisation model (Naive Bayes classifier) and testing of all ML endpoints
Week 6	d) Power BI dashboard creation: connect to MongoDB data; build spending trend, category pie, and KPI visuals e) Recharts integration in React for in-app analytics charts (line, bar, pie, area graphs) f) End-to-end system integration: connect React frontend with Node.js and Python Flask services
Week 7	a) Unit testing of all APIs and ML endpoints; User Acceptance Testing (UAT) with sample data b) Bug fixing, performance optimisation, and UI/UX improvements based on testing feedback c) Cloud deployment: React on Vercel, Node.js backend on Render, Python Flask on Railway
Week 8	a) Final project report writing: abstract, methodology, system design diagrams, test results b) Presentation preparation, demo video recording, and synopsis/documentation finalisation c) Submission of complete project report, source code, deployed URL link, and synopsis